ASSIGNMENT 2: JOANA DI ROCA

① We have i.i.d random variables $x_i \sim N(\mu=0, \sigma^2)$ and $w_i \sim \text{Unif}[-a,a]$ $\forall i=1,\dots,N$.
Value a in terms of σ if $\xi = \sum_{i=1}^{N} w_i \cdot x_i$ such that $\left.\begin{matrix} \mathbb{E}[\xi]=0 \\ \sigma_\xi = A = 1 \end{matrix}\right\}$.

We start checking the zero expected value of the neuron potential:

$$\mathbb{E}[\xi] = \mathbb{E}\left[\sum_{i=1}^{N} w_i x_i\right] = \sum_{i=1}^{N} \mathbb{E}[w_i x_i] = \sum_{i=1}^{N} \mathbb{E}[w_i]\cdot\mathbb{E}[x_i] = 0$$

(linearity of $\mathbb{E}$; independence of w_i and x_i; $\mathbb{E}[x_i]=\mu=0$ and $\mathbb{E}[w_i]=\frac{a+(-a)}{2}=0$)

Now we will find the expression for $\sigma_\xi^2 = A^2$ in terms of σ and a, and then we will impose that $\sigma_\xi^2 = 1^2 = 1$ to get a in terms of σ.

$$\sigma_\xi^2 = \text{Var}(\xi) = E(\xi^2) - \cancel{E(\xi)^2}^{\,0} = E(\xi^2) = \mathbb{E}\left[\left(\sum_{i=1}^{N} w_i x_i\right)^2\right] = \mathbb{E}\left[\sum_{i=1}^{n}\sum_{j=1}^{n} w_i w_j x_i x_j\right] =$$

$$= \sum_{i=1}^{N}\sum_{j=1}^{N} \mathbb{E}[w_i w_j x_i x_j] = \sum_{i=1}^{N} \mathbb{E}[w_i^2 x_i^2] + \sum_{i=1}^{N}\sum_{\substack{j=1 \\ j\neq i}}^{N} \mathbb{E}[w_i w_j x_i x_j] =$$

$$= \sum_{i=1}^{N} \mathbb{E}[w_i^2]\cdot\mathbb{E}[x_i^2]$$

(independence of x_i^2 and w_i^2; the cross terms $\to 0$ by independence and $\mathbb{E}[w_i]=\mathbb{E}[x_i]=0$ $\forall i$)

We need to know the value of $\mathbb{E}[w_i^2]$ and $\mathbb{E}[x_i^2]$ $\forall i=1,\dots,N$:

- $\mathbb{E}[w_i^2] = \int_{-a}^{a} w_i^2 \cdot \frac{1}{2a}\, dw_i = \frac{w_i^3}{6a}\Big|_{-a}^{a} = \frac{a^2}{3}$
- By definition of expected value we have: $\mathbb{E}[x_i^2] = \int_0^{+\infty} x_i^2\, p(X_i^2 = x)\, dx_i =$

$$= \int_0^{+\infty} x_i^2 \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{x_i^2}{2\sigma^2}} dx_i = \dots \text{TOO LONG!}$$

It's easier to find it using the variance formula:

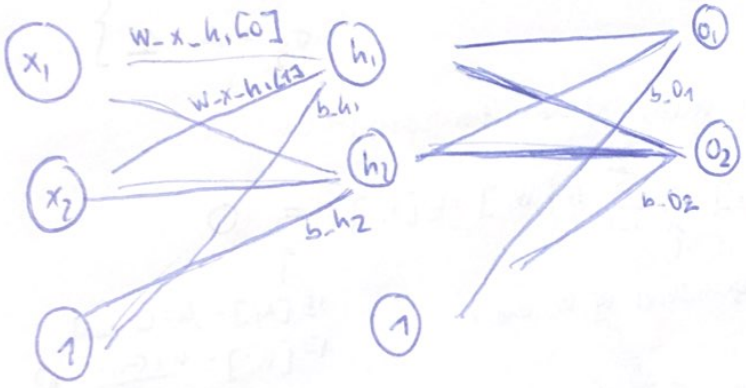
$$\text{Var}[X_i] = \mathbb{E}[x_i^2] - \mathbb{E}[x_i]^2 \Rightarrow \mathbb{E}[x_i^2] = \text{Var}(x_i) - \underbrace{E[x_i]^2}_{\mu^2=0} = \text{Var}(x_i) = \sigma^2$$

So finally, we have: $\sigma_\xi^2 = \sum_{i=1}^{N} \mathbb{E}[w_i^2]\,\mathbb{E}[x_i^2] = \sum_{i=1}^{N} \frac{a^2}{3}\cdot\sigma^2 = N\cdot\frac{a^2}{3}\cdot\sigma^2$

Imposing that $A=1$: $\sigma_\xi^2 = N\cdot\frac{a^2}{3}\cdot\sigma^2 = A^2 = 1 \Rightarrow a^2 = \frac{3}{N\cdot\sigma^2} \Rightarrow \boxed{a = \sqrt{\frac{3}{N}}\cdot\sigma}$

So the weights are uniformly distributed in the interval $\left[-\sqrt{\frac{3}{N}}\cdot\sigma,\ \sqrt{\frac{3}{N}}\cdot\sigma\right]$.
Meaning $w_i \sim \text{Unif}\left[-\sqrt{\frac{3}{N}}\cdot\sigma,\ \sqrt{\frac{3}{N}}\cdot\sigma\right]$.

(2) Drawn of the neural network, the values of weights and biases are in the jupyter notebook:



X_1	X_2	O_1	O_2
0	0	0	1
0	1	1	0
1	0	1	0
1	1	1	1